EE 446 Homework 1
Discussion Questions

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Problem 1, Part (e): Further Model Size Reduction

Comparing the results from the previous parts, the smallest model was the full-integer quantized model from part (b) at 5.76 KB, while the best-performing model was the knowledge-distilled student from part (d), which reached 100% test accuracy at 6.12 KB. Pruning alone in part (c) produced no size benefit (14.14 KB), because zeroed weights are still stored densely in a standard TFLite conversion.

The proposed strategy combines all three techniques on the smallest architecture: the distilled student model (1,027 parameters versus 3,171 for the teacher) was pruned to 70% sparsity using the same polynomial decay schedule as part (c), fine-tuned with the distillation loss, stripped of its pruning wrappers, and then converted with full INT8 quantization using a representative dataset drawn from the training features.

The resulting model is 3.71 KB with a test accuracy of 94.44%. This is roughly 35% smaller than the previous best (5.76 KB) and about 3.8× smaller than the float32 baseline. The size reduction is therefore possible, and the change that made the biggest difference was the smaller student architecture obtained through knowledge distillation, since quantization and pruning only compress the representation of whatever architecture is given to them. INT8 quantization was the second largest contributor, reducing each weight from four bytes to one. Pruning contributed essentially nothing to the file size while causing the accuracy drop: the training log shows validation accuracy falling from 100% to 92% exactly when the sparsity schedule tightened toward 70%, indicating that a 1,027-parameter model has little redundancy left to prune. The accuracy cost corresponds to three misclassified samples out of 54. Further meaningful size reduction is unlikely at this scale, because a TFLite file carries roughly 1 2 KB of fixed schema and metadata overhead, so the container rather than the weights begins to dominate the file size.

Problem 2: Exploring Edge Impulse

Question 1: Learning Blocks

The free-tier Edge Impulse account provides supervised and unsupervised learning blocks. For supervised learning, it offers Classification (for example, a keyword-spotting or gesture-recognition neural network), Regression (for example, estimating a continuous value such as fill level from sensor data), and Object Detection (for example, a FOMO model that locates objects in images). It also offers Transfer Learning blocks for images and for keyword spotting, which fine-tune a pretrained network such as MobileNet on the user's data. For unsupervised learning, it provides

Model	Size (KB)	Test accuracy
Float32 baseline (a)	14.10	0.9815
INT8 (b)	5.76	0.9815
Float16 (b)	9.00	0.9815
Dynamic range (b)	8.20	0.9815
Pruned (c)	14.14	0.9444
KD student (d)	6.12	1.0000
KD + pruning + INT8 (e)	3.71	0.9444

Table 1: Model size and test accuracy across all parts of Problem 1.

Anomaly Detection based on K-means clustering and Anomaly Detection based on a Gaussian Mixture Model, used for example to flag abnormal vibration patterns in machine monitoring. Self-supervised learning blocks are not offered in the free tier.

Question 2: Classifier Layers

The visual editor in the Classifier block of the free tier provides the following layer types. The Dense layer is a fully connected layer that combines all input features and performs the main classification computation. The 1D Convolution / Pooling layer slides filters along one axis to extract local patterns from time-series or audio spectrogram data, with pooling downsampling the result. The 2D Convolution / Pooling layer does the same over two spatial dimensions and is used for image data. The Reshape layer reorganizes the flat feature vector into the shape a convolutional layer expects. The Flatten layer converts multi-dimensional feature maps back into a vector before the dense layers. The Dropout layer randomly disables a fraction of neurons during training to reduce overfitting.

Question 3: Deployment Compression Options

Under the Deployment section, the free tier offers two model optimization choices and one compiler optimization. The model can be exported as an unoptimized float32 version or as a quantized INT8 version, which uses a representative dataset to convert weights and activations to 8-bit integers for a smaller and faster model. In addition, the EON Compiler can be enabled, which compiles the network directly to C++ source instead of using the TFLite interpreter, reducing both RAM and flash usage; Edge Impulse also provides an EON Compiler (RAM optimized) variant that trades some flash for lower RAM.

Question 4: Synthetic Data Generation

The free tier supports three input modalities for synthetic data generation. For images, it integrates an image-generation model (DALL-E) that creates labeled images from text prompts. For human speech, it uses a text-to-speech service (OpenAI TTS) to generate spoken keyword samples in different voices for keyword-spotting datasets. For general audio, it integrates a sound-effect generation service (ElevenLabs) that produces non-speech sounds such as glass breaking or dog barks. Synthetic data is important because collecting and labeling real sensor data is often the most expensive part of an embedded machine learning project; generated data lets a developer expand small datasets, cover rare or dangerous events that are hard to record, and balance underrepresented classes, which improves model robustness before any real deployment data is available.